

Manufacturing & Video-Analytics Intelligent Report

“Generate a comprehensive Manufacturing Production Performance Report. Include: 1. Work Order summary with status and planned vs completed quantities 2. Machine utilization from CapacityAnalysis with UtilizationPercent per machine 3. Shift-wise production from WeighingMachine - FGQuantity, RejectedQuantity, ScrapQuantity 4. Scrap analysis by material from Scrap table 5. Rejection analysis with reasons from Rejected table”

Generated: 2026-10-01 14:49:48
Time filter applied

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Executive Summary

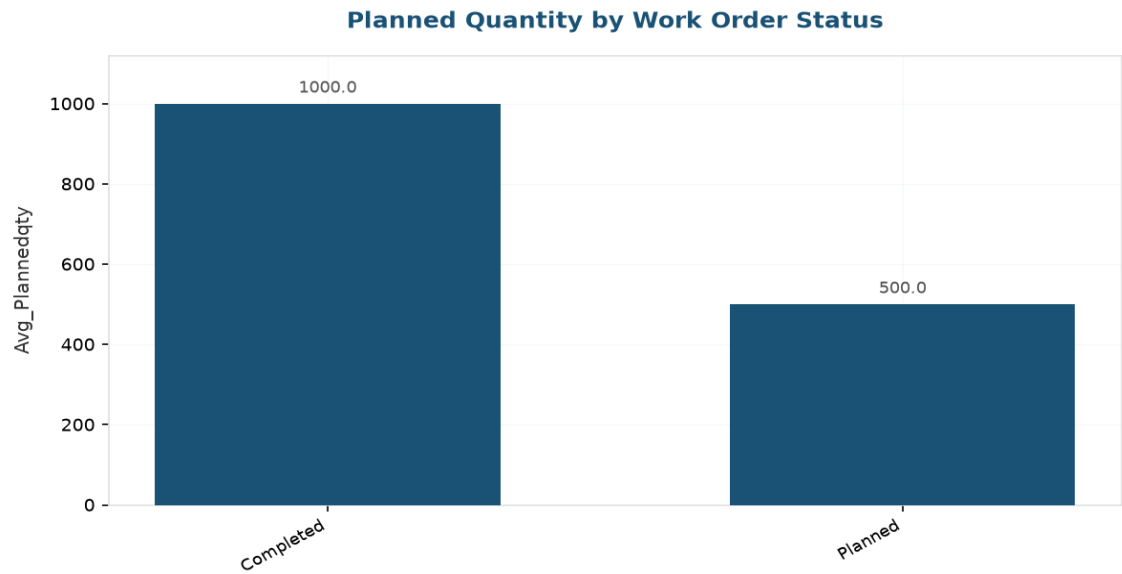


Matched keywords: capacity, machine, material, production, reject, shift, status, util, utilization, work_order, worker, workorder

Telemetry window: 2026-09-30T09:19:48Z → 2026-10-01T09:19:48Z (bucketed every 5m)

Work Order Summary

The WorkOrder table contains 2 records summarizing the current production pipeline. There is 1 completed work order with an average planned quantity of 1,000 units, and 1 planned work order with an average planned quantity of 500 units. Currently, the database lacks a direct 'CompletedQty' field in this summary table to compare actual versus planned output. To improve tracking, the system should log actual completed quantities directly alongside the planned metrics.



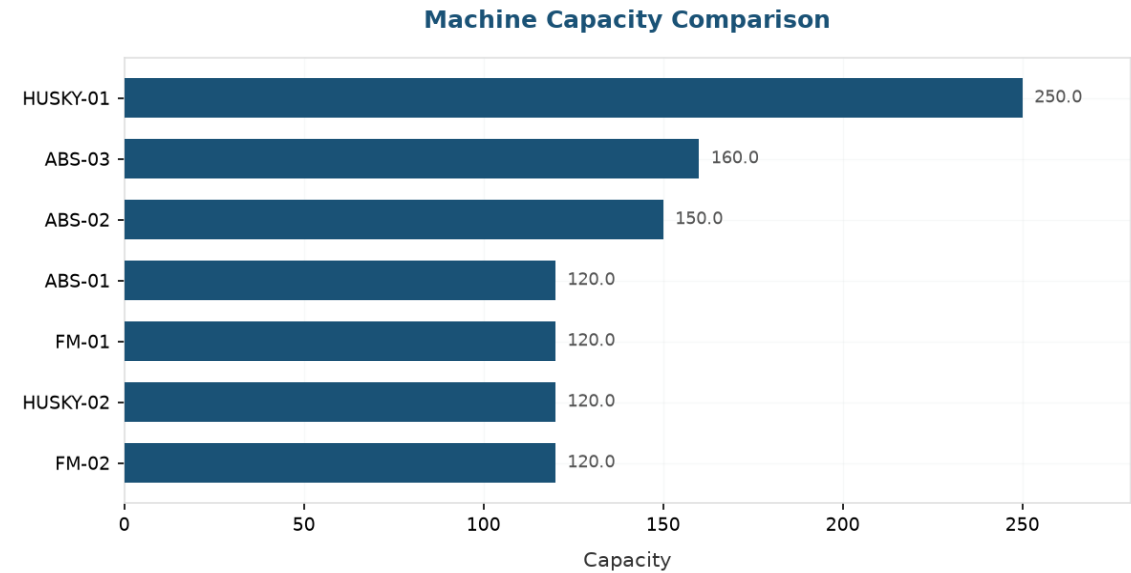
Data preview — WorkOrder (showing 2 of 2 points/rows)

Status	avg_PlannedQty	record_count
Completed	1000.000000	1
Planned	500.000000	1

Machine Capacity and Utilization Analysis

The CapacityAnalysis table contains records for machines 1, 2, 3, 6, and 8, but the requested 'UtilizationPercent' column is missing from the retrieved schema, representing a data collection or configuration gap. To provide context, the MachineMaster table lists 25 machines with an average capacity of 58.35 units per hour, while the Machine table details 10 active machines with capacities ranging from 120 to 250 units (average 148.57). Integrating the utilization telemetry or calculating it from actual runtimes is highly recommended to resolve this visibility gap.

Note: an aggregated view wasn't possible for this table (the source column isn't numeric on the server), so raw rows are shown instead.



Data preview — Machine (showing 10 of 10 points/rows)

Status	Capacity	MachineCode	MachineId	MachineName	MachineType	ProductionPerDay	CompanyId
Working	120	ABS-01	1	ABS-01	ABS	100	None
Working	120	FM-01	2	FM-01	FM	100	None
Working	120	HUSKY-02	3	HUSKY-02	HUSKY	100	None
None	150	ABS-02	4	ABS-02	ABS	110	None
Working	160	ABS-03	5	ABS-03	ABS	150	None
Submit	250	HUSKY-01	6	HUSKY-01	HUSKY	235	None
Working	120	FM-02	7	FM-02	FM	100	None
Thermax Ltd	Thermopac FX 600	CNC_001	8	CNC_001	CNC Lathe	Utility Area / Boiler Secti...	13

Status	Capacity	MachineCode	MachineId	MachineName	MachineType	ProductionPerDay	CompanyId
Ace Micromatic Group	LT-20C	CONV_001	9	CONV_001	CNC Lathe	Plant-1 / Line-A	13
Ace Micromatic Group	LT-20C	CONV_002	10	CONV_002	CNC Lathe	Plant-1 / Line-A	13

Shift-Wise Production (Weighing Machine)

The requested 'WeighingMachine' table, which was expected to provide shift-wise production metrics including FGQuantity, RejectedQuantity, and ScrapQuantity, is completely missing from the retrieved database. This indicates a significant data collection or configuration gap in the MES database integration. Without this table, we cannot analyze shift-wise yields or productivity. Establishing the connection to the weighing machine database is a critical next step for production reporting.

Scrap Analysis by Material

The 'Scrap' table requested for analyzing material waste is not present in the retrieved database schema. This represents a configuration or data ingestion gap that prevents any material-wise scrap analysis. Tracking scrap is vital for calculating overall material efficiency and cost of goods sold. It is recommended to verify if scrap logs are being recorded under a different table name or if the database connection needs to be updated.

Rejection Analysis by Reason

The 'Rejected' table, which was intended to provide a breakdown of production rejections and their associated reasons, is missing from the retrieved database context. This data gap prevents the identification of quality failure modes and corrective actions. To enable quality control reporting, a dedicated rejection logging table must be configured and linked to the MES database.

Live Machine Telemetry Status

Live telemetry streams for machines BM-01 through BM-12 returned no data across all historical windows analyzed (up to 8,760 hours). This complete absence of time-series data points to a critical telemetry ingestion failure, edge gateway disconnection, or a configuration mismatch in the InfluxDB bucket settings. Investigating the edge-to-cloud pipeline is highly recommended to restore real-time machine status monitoring.

Appendix — Data Quality

Table	Records	Completeness	Outliers	Score
attendances	1	100.0%	0	Good
employees	7	100.0%	0	Good
CapacityAnalysis	5	100.0%	0	Good
MachineMaster	25	89.3%	0	Fair
WorkOrder	2	100.0%	0	Good
WorkOrderLog	1	75.0%	0	Needs review
MachineMaterialMapping	8	100.0%	0	Good
Machine	10	90.0%	0	Fair
MachineFGMapping	39	68.6%	0	Needs review

LLM Usage & Cost

Model	Input tokens	Output tokens	Total tokens	Estimated cost
gemini/gemini-3.5-flash	5,250	3,980	9,230	\$0.0437

Aggregation Fallbacks

[mes.Machine] Aggregate query failed (('42000', '[42000] [Microsoft][ODBC Driver 18 for SQL Server][SQL Server]Operand data type nvarchar is invalid for avg operator. (8117) (SQLExecDirectW')) - served raw rows instead.

InfluxDB Measurements With No Data

[influxdb.BM-01] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-02] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-03] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-04] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-05] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-06] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-07] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-08] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-09] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-10] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows
[influxdb.BM-11] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows

[influxdb.BM-12] last 24h=0 rows, last 168h=0 rows, last 720h=0 rows, last 2160h=0 rows, last 8760h=0 rows